

ITA06 - Machine Learning

EXP 3: Demonstrate the working of the decision tree based ID3 algorithm. Use an appropriate data set for building the decision tree and apply this knowledge to classify a new sample using google colab

Aim

To implement and demonstrate the ID3 algorithm for building a Decision Tree and classify a new sample using Google Colab.

Program

```
# Import libraries

import pandas as pd

import numpy as np

from sklearn.preprocessing import LabelEncoder

from sklearn.tree import DecisionTreeClassifier, export_text

# Create Play Tennis Dataset

data = {

    'Outlook':['Sunny','Sunny','Overcast','Rain','Rain','Rain',

               'Overcast','Sunny','Sunny','Rain','Sunny',

               'Overcast','Overcast','Rain'],

    'Temperature':['Hot','Hot','Hot','Mild','Cool','Cool',

                  'Cool','Mild','Cool','Mild','Mild',

                  'Mild','Hot','Mild'],

    'Humidity':['High','High','High','High','Normal','Normal',

               'Normal','High','Normal','Normal','Normal',

               'High','Normal','High'],

    'Wind':['Weak','Strong','Weak','Weak','Weak','Strong',

            'Strong','Weak','Weak','Weak','Strong',

            'Strong','Weak','Strong'],

    'PlayTennis':['No','No','Yes','Yes','Yes','No',

                  'Yes','No','Yes','Yes','Yes',
```

```

        'Yes','Yes','No']
    }

df = pd.DataFrame(data)
print("Dataset:")
print(df)

# Encode categorical data
le = LabelEncoder()

for column in df.columns:
    df[column] = le.fit_transform(df[column])

# Features and Target
X = df.iloc[:, :-1]
y = df.iloc[:, -1]

# Build ID3 Decision Tree
clf = DecisionTreeClassifier(criterion='entropy')
clf.fit(X, y)

# Display Tree Rules
print("\nDecision Tree Rules:")
print(export_text(clf, feature_names=list(X.columns)))

# New Sample
# Outlook=Sunny, Temperature=Cool,
# Humidity=High, Wind=Strong
new_sample = [[2,0,0,0]]

prediction = clf.predict(new_sample)

if prediction[0] == 1:
    print("\nPrediction: Play Tennis = Yes")
else:
    print("\nPrediction: Play Tennis = No")

```

OUT PUT :

```

... Dataset:
      Outlook Temperature Humidity   Wind PlayTennis
0      Sunny          Hot      High   Weak         No
1      Sunny          Hot      High Strong         No
2  Overcast          Hot      High   Weak         Yes
3       Rain      Mild      High   Weak         Yes
4       Rain      Cool   Normal   Weak         Yes
5       Rain      Cool   Normal Strong         No
6  Overcast      Cool   Normal Strong         Yes
7      Sunny      Mild      High   Weak         No
8      Sunny      Cool   Normal   Weak         Yes
9       Rain      Mild   Normal   Weak         Yes
10     Sunny      Mild   Normal Strong         Yes
11  Overcast      Mild      High Strong         Yes
12  Overcast      Hot    Normal   Weak         Yes
13     Rain      Mild      High Strong         No

```

```

... Decision Tree Rules:
|--- Outlook <= 0.50
|   |--- class: 1
|--- Outlook > 0.50
|   |--- Humidity <= 0.50
|       |--- Outlook <= 1.50
|           |--- Wind <= 0.50
|               |--- class: 0
|               |--- Wind > 0.50
|                   |--- class: 1
|           |--- Outlook > 1.50
|               |--- class: 0
|       |--- Humidity > 0.50
|           |--- Wind <= 0.50
|               |--- Temperature <= 1.00
|                   |--- class: 0
|                   |--- Temperature > 1.00
|                       |--- class: 1
|           |--- Wind > 0.50
|               |--- class: 1

```

Prediction: Play Tennis = No

/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names, so DecisionTreeClassifier will assume that each feature present in the data is a feature name.

Result

The ID3 Decision Tree was successfully constructed using the Play Tennis dataset. The trained model classified the new sample and predicted whether tennis can be played based on weather conditions.